Zoology Legends

ZOO503_Current_Final_TERM_

Important MCQS_2025

BY Muhammad Zaman

Admin of Zoology LEGENDs

- ✓ **Placoderm** are (oldest jaw bearing fishes) such as **Coccosteus**
- ✓ **Early Primates** □ One of the oldest of the modern placental mammal groups
- ✓ Which are threatened animal Of Pakistan Himalayan brown bear , Indian wolf, Asiatic black bear, (**All**)
- ✓ From the Greek Anthropos and Kainos (**Human being and new**)
- ✓ Newland and it's total area(**270,000 km2**)
- ✓ The acme of group gastropods with the radiation of the siphonal carnivorous neogastropods was in? (**Cenozoic**)
- ✓ Water vascular system is present in?(**Echinoderms**)

Zoology Legends

- ✓ The type of jawless fishes, ostracoderm were jawless, generally armored and had their hey day in the?(**Devonion**)
- ✓ Amniotes radiated in? (**Late Carboniferous**)
- ✓ Early crocodilian was more diverse in? **(both jurrassic and Cretaceous**)
- ✓ The oldest human was found is approximately lived ago? (**6Ma**)
- ✓ There are main extinction in brachiopod? (**Five**)
- ✓ German paleontologist established a classification action of trace fossils based on behavior? (**Adolf Seilacher**).
- ✓ Biography is an interrogative field of inquiry that unites concepts and information from following branches of science except?(**physics**)
- ✓ Which scientist translated Cuvier's idea into English in 1813? (**Robert Jameson**)
- ✓ Biography of the ocean is called? (**Marine biogeography**).

Zoology Legends

- ✓ The sum of all the niche requirements under ideal condition when the species is given unimpeded access to resources called? (**Fundamental nich)** ya 2 time pouchi the.
- ✓ Is a term sometimes used to describe the wave of the biological invasion that has occurred in the wake of human invasion by either accidental transport or deliberate? (**Ecological Imperialism**).
- ✓ Who make world map indicating four floral regions for the first time was? (**Adolf Engler**)
- ✓ **19.90%** of brachiopod specie extinct during extinction? (**End . Permian**)
- ✓ Certain species play key role in the ecosystem and when they lost there is a sudden drop in the capacity of the ecosystem of the function?(**Rivet hypothesis**)
- ✓ In term of diversification of life during the early Cambrian was an important part of ecosystem?(**Predations**).
- ✓ Oldest known bird found in. (**Upper Jurassic sediment**).

Zoology Legends

- ✓ Brrig distinguish…. zoogeographical region (**23**).
- ✓ All of the following are part of gondwana except (**North America**)
- ✓ Trade Wich found in both Northern and southern hemispheres in the region of equator is know as (**Intertropical convergence zone ITCZ**)
- ✓ modiflication of aparin ….Haldane biochemical models is (**Hydrothermal model**).
- ✓ Parazoan body plan is found in (**Sponge**)
- ✓ homo habilis species liver nearly.(**2ma**)
- ✓ Scala Naturae is? **the ladder of life**
- ✓ Rohde and Mullet 2005 argued that there is a periodicity in mass extinction? **62mya**
- ✓ The acme of group gastropoda with the radiation of the siphonal carnivorous neogastropods was in? **Cenozoic**
- ✓ Water vascular system is present in? **Echinoderm**

Zoology Legends

- ✓ The type of jawless fishes, ostracoderm were jawless, generally armored and had their hey day in the? **Devonian**
- ✓ Amniotes radiated in? **Carboniferous**
- ✓ Early crocodilian was more diverse in? **Cretaceous and jurrassic**
- ✓ The oldest human was found is approximately lived ago? **3.75**
- ✓ There are main extinction in brachiopod? **5 mass extinction**
- ✓ German paleontologist established a classification action of trace fossils based on behavior? **Adolf seilacher**
- ✓ Biography is an introgrative field of inquiry that unites concepts and information from following branches of science except? **Ecology geology**
- ✓ Which scientist translated Cuvier's idea into English in 1813? **Robert jamson**
- ✓ Biography of the ocean is called? **Marine**

Zoology Legends

- ✓ The sum of all the niche requirements under ideal condition when the species is given unimpeded access to resources called? **Fundamentals niche**
- ✓ Is a term sometimes used to describe the wave of the biological invasion that has occured in the wake of human invasion by either accidental transport or deliberate? **Ecological imperialism**
- ✓ Among gastropods exogastric condition show that? **Evolution**
- ✓ Who make world map indicating four floral regions for the first time was? **Aldof engler**
- ✓ 90% of brachiopod specie extinct during extinction? **End Permian**
- ✓ Huronian glaciation period was? **2.4_2.1**
- ✓ Cnidarians have life strategies? **Two**
- ✓ Lagerstatten is a? **Fossil**
- ✓ The geological time shows climate zone? **5**

Zoology Legends

- ✓ The sum of all the niche requirements under ideal condition when the species is given unimpeded access the resource called? **Fundamental Niche**
- ✓ Geological studies of new Caledonia show that the island was submerged for long period in the Paleocene Eocene so that the island was not available for colorization until about? **37 mya**
- ✓ The ancestors of the mammoths and modern Asian elephant probably evolved in Africa 4 to 5 mya and migrated back into Eurasia **2 million years** later
- ✓ Certain species play key role in the ecosystem and when they lost there is a sudden drop in the capacity of the ecosystem of the function? **Rivert hypotheses**
- ✓ In term of diversification of life during the early Cambrian was an important part of ecosystem? **Predation**
- ✓ Newzeland distance..... kilometers.(**270 000 km2**)
- ✓ Which one is not Trace Fossi (**Teeth**)

Zoology Legends

- ✓ Oldest multicellular eukaryotes evolved in..(**1200Ma**)
- ✓ Observations of the appearance and disappearance of species. (**Pattern**)
- ✓ The substantial skeleton preanthropus affrain live approximately...(**3.2 Ma**)
- ✓ (**New National Museum of Natural History**) became a powerhouse of ideas and debate in Europe.
- ✓ The sum of all niche requirements under ideal conditions when the species is given unimpeded access to resout is called..(**fundamental niche**)
- ✓ The brain size of lucy was? (**415 cm3**)
- ✓ The threatened species of Pakistan (**All**)
- ✓ Protected areas that cover of terrestrial land area. (**14.6**%)
- ✓ Richness of species, particularly plants, (**increases with altitude**)
- ✓ Milk major constituent is the (**both options were correct**)
- ✓ human genetic makeup is shared with the chimpanzee..(**99%**)

Zoology Legends

- ✓ Kleptoparasitism role is to Dive for their food and bring vegetable matter to the surface from greater depths
- ✓ Kleptoparasitism role an effective way of **widening the niche of the gadwall**
- ✓ Kleptoparasitism: A very widespread duck in the gadwall (**Anas strepera**)
- ✓ Group Gastropoda with the radiation of the siphonal carnivorous
- ✓ neogastropods..(**Cenozoic**)
- ✓ How many zoogeographical region distinguished by Brigg... (**23**).
- ✓ Argued that all of the areas connected by one of these tracks had originally formed a single, continuous area (**Panbiogeography**)
- ✓ Chondricthyans are fishes...(**Cartilaginous**)
- ✓ The early Devonian period starts.... (**380Mya**)
- ✓ All of the following are included in ecologicalbiogeography except……(**Altitude**)

Zoology Legends

- ✓ Focus on **evolution topic + climate changes+ importance each topic**
- ✓ Define waly thy
- ✓ Ma / myc years waly both ziyada thy
- ✓ Scientist waly bhi thy kuch
- ✓ or handouts ko smjh kr prhy achy sy
- ✓ **Ostracoderms**: Were jawless, generally armored, Had their heyday in the -------**Devonian**
- ✓ The Huronian glaciation period occurred approximately **2.4 to 2.1** billion years ago during the Proterozoic eon.
- ✓ Geological studies show that the island was submerged for long periods in the Paleocene– Eocene, so that the island was not available for colonization until about ----------**37 mya**
- ✓ All of the following are part of gondwana except (**North America)**

ZOOLOGY LEGENDS

Zoology Legends

- ✓ **Panbiogeography** Argued that all of the areas connected by one of these tracks had originally formed a single, continuous area that was inhabited by the groups concerned.
- ✓ Early Devonian period starts about__**380___mya**.
- ✓ The belly-footed mullusks are ___ **Gastropods**
- ✓ Are the observation of the appearance and disappearance of species?___ **Patterns**
- ✓ Geological studies show that the island was submerged for long periods in the paleocene– Eocene, so that the island was not available for colonization until about **37mya**.
- ✓ The evolution by natural selection model was presented by **_darwin**
- ✓ About **3 mya**, an increase in aridity, related to the uplift of the Himalayan Mountains to new
- ✓ heights, led to the northern parts of the **Indian** subcontinent becoming much drier and also led to an increase in the numbers of

grazing mammals such as **horses, antelope and camels, and of elephant**

- ✓ **Allopatric speciation** Also referred to as **geographic speciation**, is a mode of speciation that occurs when biological populations of the same species become isolated from each other.
- ✓ evolution waly bhi bohat
- ✓ The demands that an organism places on its environment in terms of physical and chemical conditions, space and food supply is called **niches**
- ✓ **Biogeography** is the study of the distribution of species and ecosystems in geographic space and through geological time.
- ✓ In grassland to species of a sparrow Sparrow A and Sparrow B both Sparrow eat seed in same habitat. Suddenly reduce in seed production. What type of statement regarding this is correct? (**Interspecific competition: two different species compete for resources in their habitat.)**

Zoology Legends

- ✓ ISI Tarah if you want a hypothesis hypothesis be py research karna chahte hain. To next step kya hoga scientific experiment mein iska confusing se statements the. (**design and perform an experiment**)
- ✓ The Spirallians are morphologically diversed clade of protostome animals including all of the following (**Molluscs, Annelids, Platyhelminths**) except……..**Cnidarians**
- ✓ Early life was mainly affected by ……..**predation**
- ✓ Amphibians and reptiles evolved in …..**Triassic**
- ✓ Lucy fossil old myb…. **3.2 million years**
- ✓ Reptilian ancestors of mammals had… **synapsid skull**
- ✓ The most advance mollusks are……..**Cephalopods**.
- ✓ The oldest human was found is approximately lived __ago……**6Mya**
- ✓ All of the following part of Gondwana Except……..**North America**

ZOOLOGY LEGENDS

Zoology Legends

- ✓ The oldest substantial skeleton of praeanthropus afarensis lived approximately ----------ago…….**3.2 Mya**
- ✓ Archaeopetyx is true missing link between…….**Dinosaurs and Birds**
- ✓ Rohde and Muller (**2005**) argued that there is__ periodicity in mass extinctions …….**62 myr**
- ✓ What period of time has been suggested to be known as the '**Anthropocene**'..,..... current period
- ✓ How many of the 75 different evolutionary lineages of amphibian and reptile species are related to South American taxa?.....**49**
- ✓ **Def. Of ecology**..... A group of species that play the same role in a community.
- ✓ What is **biogeography**? The study of the distribution of species and ecosystems in geographic space and through geological time .
- ✓ By product of bivalves is .....**pearl**
- ✓ What was the age of the human fossils announced by two French teams in 2001 and 2002? **6Mya**

Zoology Legends

- ✓ What limits a species' distribution? **Both physical and biotic factors**
- ✓ Who emphasized the importance of competition in ecological biogeography? **Candolle**.
- ✓ Jo evolutionary Eras hy (**jurassic, devonian** etc.) Un myn sy bhht Mcqs thy koi 8, 10 shyd .
- ✓ Some of the Mcqs were (**mya [million years ago**]) type.
- ✓ What is **zooniverse**? refers to an online platform that allows people to participate in scientific research by assisting researchers in analyzing data
- ✓ After which mass extinctions event did diapsid reptiles diversity in the Triassic.....? **end-Permian**
- ✓ Fossil in fold belt are .....? Example is the "**Delabole butterfly**" In fact, the wide-hinged fossils are "**Spiriferide brachiopods**"
- ✓ **FOLD BELTS:** A fold and thrust belt is a series of mountainous foothills adjacent to an orogenic belt, which forms due to contractional tectonics.

Zoology Legends

- ✓ Fossils in fold belts can be of great value to **structural geologists** in understanding the rates and timing of tectonic events.
- ✓ The ancestors of the mammoths and modern Asian elephant probably evolved in Africa 4 to 5 mya and migrated back into **Eurasia 2 million** years later.
- ✓ **Charles Darwin** (1809–1882), developed the theory of evolution by natural selection in the **1830s**
- ✓ which protist is responsible for causing malaria in human ?
- ✓ Amazon rainforest is primarily located in which continent ?
- ✓ what is the definition of migration
- ✓ what is the primary factor driving allopatric speciation?
- ✓ **1300** native species of freshwater or terrestrial vertebrate in the islands, of which **75%** are endemic
- ✓ Where did the extinction megafauna begins ? – **first in Australia**, then in Eurasia and finally in North America

ZOOLOGY LEGENDS

Zoology Legends

- ✓ **Bioerosion**: The removal of skeletal materials by boring organisms such as sponges, algae and bivalves.
- ✓ Shells, bones and wood may undergo **bioerosion**
- ✓ The main characteristics porifera…. **Unique porous structure**.
- ✓ Most gastropods undergo **torsion** in early life; they have a **single shell**, often coiled
- ✓ Trochophore larva in **mollusks and the polychaetes**
- ✓ Resemblance between humans and eutheria is **nourishing by placenta**.
- ✓ One of the oldest of the modern placental mammal groups is **primates**.
- ✓ The threatened species of Pakistan: **Include Asiatic Black bear, Himalayan brown bear, Indian Wolf (All)**
- ✓ New global biogeographical classification system for coastal and shelf areas known as the **Marine Ecoregions of the world and MEOW.**

Zoology Legends

- ✓ Protected areas that cover **14.6%** of terrestrial land area.
- ✓ Euthycarcinoids member of: An enigmatic **group of arthropods** with antennae and mandibles. In phylogeny somewhere near the **origin of myriapods** and insects
- ✓ "Belly-footed" mollusks are **gastropods**.
- ✓ First vertebrates had a **cartilaginous** skeleton.
- ✓ Croizat published his ideas in the **1950s and 1960s**, his major presentation being his **1958** book **Panbiogeography**.
- ✓ Buffon was a **naturalist**
- ✓ Two chapters in On the Origin of Species were devoted to **geographical distribution**.
- ✓ **Biogeography** is the study of the distribution of species and ecosystems in geographic space and through geological time.
- ✓ Island biotas (**faunas and floras**) are often diverse, with many endemic species and commonly with evidence that these species came originally from one or more source continents.

Zoology Legends

- ✓ Only the rats spread naturally as far as **Australia**, **50 species** now form **50%** of the diversity of the Australian land mammal fauna.
- ✓ Richness of species, particularly plants, increases with **altitude**, reaches highest peak.
- ✓ **Fundamental niche**, which is the sum of all the niche requirements under ideal conditions when the species is given unimpeded access to resources.
- ✓ When a species is prevented from occupying an area by the presence of another species in this way, it is termed **competitive exclusion**.
- ✓ The simplest influences that **predators** might have is to **eliminate** species by eating them.
- ✓ **Ecological imperialism** is a term sometimes used to describe the wave of biological invasions that has occurred in the wake of human invasions.
- ✓ In the middle of the Palaeozoic, **400 mya**, a large landmass that we call **Euramerica**.

Zoology Legends

- ✓ The **equatorial regions** are the areas in which **highest productivity** is possible because of the prevailing climate, which is hot, wet and relatively free from seasonal variation.
- ✓ **Afrotheria** are the ancestors of the mammoths and modern Asian elephant.
- ✓ If there is no change in proportions during growth, the feature is said to show **isometric ("same measure**") growth
- ✓ The ratio of eye diameter to body length diminishes as the animal approaches adulthood. This is an example of **allometric ("different measure") ** growth.
- ✓ Marine faunas: wide expanses of deep ocean, swift **ocean currents** or land.
- ✓ Although the polar regions receive less energy from the sun than the equatorial regions, they have been supplied with warmth by a free circulation of **ocean currents** through most of the world's history

Zoology Legends

- ✓ The oldest locomotory trace fossils are from about **550 Ma** northwest Russia
- ✓ **Trace fossils**: the remains of the activity of ancient organisms, such as burrows and tracks.
- ✓ Trace fossils may be produced within a **sedimentary layer**, or on the surface
- ✓ The study of trace fossils, often called **ichnology**
- ✓ The most advanced mollusks are -------- **Cephalopods**
- ✓ **Aplacophorans** A group of shell-less mollusks
- ✓ may be external (exoskeleton) in the case of most **invertebrates**
- ✓ Internal (endoskeleton) structures, in the case of a few **mollusks**
- ✓ **Trochophore larva**: mollusks and the polychaetes
- ✓ Shelled veliger: **mollusks**
- ✓ The Phylum Mollusca can be traced back to at least the **Late Precambrian**
- ✓ One McQ related to Antarctica freezing out.

Zoology Legends

- ✓ By the Early Oligocene, around **35 mya**, it had separated sufficiently far from Antarctica for the deep water Antarctic Circumpolar Current of cooler water and associated westerly winds to become established
- ✓ During the most recent Ice Age, the movement of Antarctica into its position over the South Pole led to the development of a Southern **Polar Ice Cap 42 mya**.
- ✓ Falling global temperature then led to then development of the Greenland Ice Cap **around 34 mya**
- ✓ Shallow-water coral reefs in zone extending 30° north and **south of equator**. 30m or 18 °C
- ✓ The 'trade winds', found in both the Northern and Southern Hemispheres, meet in the region of the equator, and this is known as **the intertropical convergence zone (ITCZ**).
- ✓ The **equatorial regions** are the areas in which highest productivity is possible because of the prevailing climate, which is hot, wet and relatively free from seasonal variation.

Zoology Legends

- ✓ New Guinea contributes only 3% of the world's land area, it contains some **10%** of its species of terrestrial organism.
- ✓ All the land surface area of the hotspots together comprise only about **1.4% of** the Earth's terrestrial total
- ✓ The **Gymnosperms**: Seed-bearing plants radiated in several phases. During the Carboniferous-Permian (**medullosans, cordaites, cycads**) and Mesozoic (conifers, ginkgos, bennettitaleans, gnetales).
- ✓ The **Angiosperms**: Flowering plants radiated dramatically during the Cretaceous, fully enclosed and protected seeds, flowers and double fertilization.
- ✓ Angiosperms have always been most abundant in the low-latitude (**tropical**) regions
- ✓ Oldest multicellular eukaryotes: reported from rocks **1200 Ma** in Canada
- ✓ These red algae were one of nearly **20 multicellular eukaryote** lineages

ZOOLOGY LEGENDS

Zoology Legends

- ✓ One of the oldest multicellular organisms/eukaryote is **Bangiomorpha** 1.2 Ga
- ✓ One McQ related to colonization.
- ✓ Geological studies show that the island was submerged for long periods in the Paleocene–Eocene, so that the island was not available for colonization until **about 37 mya**.
- ✓ Biogeographers have investigated the history of the angiosperm family Sapotaceae. Resulted from **nine colonizations** of the island between 33 and 4.2 mya, dates that are about 40 million years after the rifting of New Caledonia from Australia.
- ✓ The threatened species of Pakistan Include **Asiatic Black bear, Himalayan brown bear, Indian Wolf (All)**
- ✓ Only the rats spread naturally as far as Australia, 50 species now form **50% of the diversity** of the Australian land mammal fauna.
- ✓ Richness of species, particularly plants, increases with **high altitude**, reaches highest peak.

Zoology Legends

- ✓ In the middle of the Palaeozoic, 400 mya, a large landmass that we call **Euramerica**.
- ✓ The evolution by natural selection model was presented by **Darwin**
- ✓ Human genetic makeup is shared with the chimpanzee. **(99%)**
- ✓ What period of time has been suggested to be known as the'**Anthropocene'** current period
- ✓ Who emphasized the importance of competition in ecological biogeography? **Candolle**
- ✓ **Archaeopetyx** is true missing link between Reptiles and Birds
- ✓ **Stenolaemata:** Dominated Paleozoic bryozoan faunas
- ✓ Protected areas that cover of terrestrial land area. **(14.6%)**
- ✓ New Zealand distance: (**270,000 km²)**
- ✓ Observations of the appearance and disappearance of species: (**Pattern**)
- ✓ The substantial skeleton of Preanthropus afarensis lived approximately: (**3.2 Ma**)

Zoology Legends

- ✓ The sum of all niche requirements under ideal conditions when the species is given unimpeded access to resources: (**Fundamental niche**)
- ✓ Richness of species, particularly plants, (**increases with altitude**)
- ✓ Milk major constituent: (**Both options were correct**)
- ✓ **Kleptoparasitism** role: (Not specified, but involves **stealing food from other anima**ls)
- ✓ Group Gastropoda with the radiation of the siphonal carnivorous neogastropods: (**Cenozoic**)
- ✓ How many zoogeographical regions distinguished by Brigg: (**23**)
- ✓ **Panbiogeography**: Argued that all of the areas connected by one of these tracks had originally formed a single, continuous area.
- ✓ Chondrichthyans are fishes: (**Cartilaginous**)
- ✓ The early Devonian period starts: (**380 Mya**)
- ✓ All of the following are included in ecological biogeography except: (**Altitude**)

Zoology Legends

- ✓ Examines how plants and animals vary in their physiological processes in response to the environment: (**Ecophysiology**)
- ✓ Icehouse and greenhouse conditions: (**Occurred 5 times**)
- ✓ Share many anatomic characters and **>94%** of their DNA is identical
- ✓ Human Closely related to the great apes, which include the **orangutan Pongo**, and especially to the gorilla
- ✓ The separation of the human ancestral branch (the **hominins**) from the great apes (the two groups are known jointly as hominids) taken place about **7 mya.**
- ✓ It is very likely that **hominids** (which include the ancestors of apes and humans) followed similar changing patterns of distribution in the **Miocene**.
- ✓ speciation se MCQs th bht zyada mold kr k diye hue th
- ✓ evolution se bhi questions th wo long statements thn
- ✓ The Spirallians are morphologically diversed clade of protostome animals including all of the following except…….. **Cnidarians**

Zoology Legends

- ✓ Early life was mainly affected by …..….. **predation**
- ✓ Amphibians and reptiles evolved in ….. **Triassic**
- ✓ Lucy fossil old myb**…. 3.2 million years**
- ✓ Reptilian ancestors of mammals had… **synapsid skull**
- ✓ The most advance mollusks are…….. **Cephalopods**
- ✓ The oldest human was found is approximately lived ___ago……. **6Mya**
- ✓ All of the following part of Gondwana Except…….. **North America**
- ✓ The oldest substantial skeleton of praeanthropus afarensis lived approximately ----------ago……. **3.2 Mya**
- ✓ Archaeopetyx is true missing link between……. **Dinosaurs and Birds**
- ✓ Rohde and Muller (2005) argued that there is__ periodicity in mass extinctions **……. 62 myr**

ZOOLOGY LEGENDS

Zoology Legends

- ✓ What period of time has been suggested to be known as the 'Anthropocene'..,..... **current period**
- ✓ How many of the 75 different evolutionary lineages of amphibian and reptile species are related to South American taxa?..... **49**
- ✓ **Def. Of ecology**...,..... A group of species that play the same role in a community.
- ✓ What is **biogeography**? The study of the distribution of species and ecosystems in geographic space and through geological time .
- ✓ By product of bivalves is ..... **pearl**
- ✓ What was the age of the human fossils announced by two French teams in 2001 and 2002? **6Mya**
- ✓ What limits a species' distribution? **Both physical and biotic factors**.
- ✓ Who emphasized the importance of competition in ecological biogeography? **Candolle**.
- ✓ Newzeland distance..... kilometers**.(270 000 km2**)
- ✓ Which one is not Trace Fossi (**Teeth**) Teeth

Zoology Legends

- ✓ Oldest multicellular eukaryotes evolved in..(**1200Ma**)
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- ✓ (**New National Museum of Natural History**) became a powerhouse of ideas and debate in Europe.
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- ✓ The brain size of lucy was? (**415cm3**)
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- ✓ Milk major constituent is the (**both options were correct**)
- ✓ human genetic makeup is shared with the chimpanzee..(99%)

Zoology Legends

- ✓ Group Gastropoda with the radiation of the siphonal carnivorous neogastropods..(**Cenozoic**)
- ✓ How many zoogeographical regiondistinguished by Brigg... (**23**).
- ✓ Argued that all of the areas connected by one of these tracks had
- ✓ originally formed a single, continuous area (**Panbiogeography**)
- ✓ Chondricthyans are fishes...(**Cartilaginous**)
- ✓ The early Devonian period starts.... (**380Mya**)
- ✓ All of the following are included in ecological biogeography except……(**Altitude**)
- ✓ Water vascular system present in -

  Answer: **Echinoderms**
- ✓ Early crocodile diverse in -

  Answer: **Africa**
- ✓ Oldest known bird fossil -

  Answer: **Archaeopteryx**
- ✓ Basal part of metazoa life made from -

Zoology Legends

Answer: **Sponges**

✓ Belly footed molluscus -

Answer: **Gastropods**

✓ Which continent was focus of human evolution -

Answer: Africa

✓ Continent fragment move in 1912 by -

Answer: **Alfred Wegener**

✓ Species richness at altitude -

Answer: **Increases**

✓ Milk contain -

Answer: **Lactose**

✓ Theory of evolution by -

Answer: **Charles Darwin**

✓ _____is a term sometimes used to describe the wave of biological invasion that has occurred____**Ecological imperialism**

Zoology Legends

- ✓ Who make world map indicating four floral regions for the first time was?---------- **Adolf Engler**
- ✓ Geological studies show that the island was submerged for long periods in the Paleocene– Eocene, so that the island was not available for colonization until about **---------37 mya**
- ✓ 4.90% of brachiopod specie extinct during extinction? (**End . Permian)**
- ✓ homo habilis species liver nearly.(**2ma**)
- ✓ The oldest substantial skeletons, of **Praeanthropus** afarensis**, 3.2 Ma**
- ✓ The '**trade winds'**, found in both the Northern and Southern Hemispheres, meet in the region of the equator, and this is known as the **intertropical convergence zone (ITCZ)**.
- ✓ Jaws among fishes evolved during early **Ordovician period**.
- ✓ Species rashness, particularly plants, **increases with altitude**, reaches a peak and then declines again at very high altitudes
- ✓ major constituent of milk ____**carbohydrates lactose**

Zoology Legends

- ✓ mostly MCQs were scenario based, make strategies, select best hypothesis according to given situation and from different eras...
- ✓ Fossils from the deformed zones of mountain belts. They are usually poorly preserved. **Metamorphosed and tectonized**
- ✓ Fossils in fold belt.....**Cliff**, ridge, steep (Not sure)
- ✓ Which one is naturalist.... **Buffon**, Wallace, Darwin
- ✓ **Panbiogeography**: Argued that all of the areas connected by one of these tracks had originally formed a single, continuous
- ✓ The three great evolutionary faunas of Phanerozoic, Cambrian, **Paleozoic** modern Developed during last **550 myr**
- ✓ The majority of "**Paleozoic**" taxa were derived from **Cambrian** stocks
- ✓ ..... Is start of life...Mitosis, replication, **abiogenesis**
- ✓ Georges Cuvier The science of **comparative anatomy**
- ✓ **Reality of extinction**: French natural scientist Georges Cuvier

Zoology Legends

- ✓ % age of Sickle cell anaemia in Africa.. 5%, **more than 20%,** less than15%,50%

Other Mid Term Papers 2025

- ✓ Earth has oscillated at least.....times between icehouse and greenhouse conditions. (**five**)
- ✓ Cretaceous – Tertiary exticiton------ (**Mesozoic , Cenozoic**)
- ✓ Order and succession of rocks is called...(**lithostratography**)
- ✓ Isthmus of panama separated...(**Atlantic and Pacific ocean**)
- ✓ Oparin halin biochemical model is... (**Hydrothermal**)
- ✓ Chromistan includes all except...(**protozoa**)
- ✓ Planula is larval stage of the_____(**cnidarians**)
- ✓ The composition and structure of fossil communities._ (**paleasynecology**)
- ✓ Bowerbankia is a genus of____ (**Bryozoans**)

Zoology Legends

- ✓ Aplacophorans are ____ (**shell-less mollusks**)
- ✓ Chelicerates are diversed and heterogeneous which include__ {**All of the above** (Mites, Scorpions, Spiders)
- ✓ Trilobites dominated assemblages in England and Wales area were described by _(**William Smith**)
- ✓ current extinction crises is sometime called (**6th extinction**)
- ✓ ___Biomarker indicates that eukaryotes were present around at least **2.7 Ga. (Lipid**)
- ✓ Shallow-water coral reefs exist less than -------(**18C)**
- ✓ The ----- are called belly- footed mollusks. (**Gastropods**)
- ✓ Evolution by natural selection model was presented by _ (**Charles Darwin**)
- ✓ When local ecological conditions affect the form of an organism during its life time and consequently fossil is called …….(**Ecophenotypic Effects**)
- ✓ progressionism also called... (**orthogenesis**)

Zoology Legends

- ✓ scientia mean... (**Knowledge**)
- ✓ “Lagerstätten” is a..... (**layer of earth**)
- ✓ Scala Naturae means**.... a)ladder of life** ✓
- ✓ Diatom radiolarian..... **Silica**
- ✓ Insects population.... **10 million**
- ✓ Crocodile more diverse in..... **Jurassic and cretaceous**
- ✓ Stenolaemata are a class of..... **bryozoan**
- ✓ Who purposed natural selection..... **Charles Darwin**
- ✓ Phylum platyhelminthes symmetry..... **Bilateral**
- ✓ Signor lipps effect..... **fossilization**
- ✓ What are fossil..... **remains of plant and animal body**

ZOOLOGY
LEGENDS

- ✓ **Archaeopteryx** is a true missing link between

Zoology Legends

- ✓ Octopus have _____ **lost skeleton**
- ✓ Who have pores in body ___ **sponges**
- ✓ Fossil information helps in identifying the level of _____ ***maturity of rocks.***
- ✓ **George Cuvier** is known as father of
- ✓ Law of superposition was formulated by which **danish geologist**
- ✓ Who divided the world **into 6** geographical regions
- ✓ Assessment of rare earth metals such as _____ to help pinpoint the time of death. "**Scandium/Uranium**"
- ✓ Science mean ***Knowledge***
- ✓ Father of comparative anatomy ***Curvier***
- ✓ Life on Earth ***4 Billion***
- ✓ Mass extinction ***6 time***
- ✓ Exoskeleton of arthropods made up of ***chitin***
- ✓ Head fused with thorax ***cephalothorax***
- ✓ Bivalved was named by ***Linnaeus***

Zoology Legends

- ✓ Groups of conodont *3*
- ✓ Human arose *6-8mya*
- ✓ Insects first appeared *early devonian*
- ✓ Signor lipps effect *straightography*
- ✓ Bivalves is characterized by *All*
- ✓ Insect Population *10 million*
- ✓ Animal feed on died plants and animals *Scavenging*
- ✓ Adapted wala tha defination *Natural Selection*
- ✓ Structure and composition of fossil communities *Paleosynecology*
- ✓ Membrane enclose the embryo *Amnion*
- ✓ *Filter feeder* wala
- ✓ Iso metric & allometric *growth*
- ✓ The complete or partial remain of plant & animal *body fossil*
- ✓ Mass extinction on earth *6 Times*
- ✓ Crocidile more diverse in *Jurassic & Cretaceous*
- ✓ Shallow-water coral reefs in zone.... *18 °C*

Zoology Legends

- ✓ “Archaeopteryx”, true “missing link” between.... ***Birds and Reptiles***
- ✓ First living organisms..... ***Precambrianms***
- ✓ The Parazoan Body Plan..... ***metazoans***
- ✓ Aplacophorans and shelled mollusks shared a common ancestor probably during
- ✓ the.... ***Late Precambrian***
- ✓ Difference between Bed and Lithostratigraphy.... ***bed the smallest***
- ✓ Which of the following is Spiralians.....***Molluscs***
- ✓ Rostroconchs is.... ***Class Molluscs***
- ✓ Eurypterids found.... ***during the Silurian*** *and* ***Devonian***
- ✓ Arthropods characteristics..... ***segmented body plan with jointed appendages***
- ✓ Chelicerates are diversed and heterogeneous which includes…... ***Mites, Scorpions, Spiders***

Zoology Legends

- ✓ Cephalopods.... ***had a Head***
- ✓ When local ecological conditions affect the form of an organism during its life time and consequently fossil is called …….(**Ecophenotypic Effects**)
- ✓ Evolution by natural selection model was presented by _____ (**Charles Darwin)**
- ✓ __Biomarker indicates that eukaryotes were present around at least **2.7 Ga….(Lipid**)
- ✓ Jawless fishes that lived in ----------period (**Devonian**)
- ✓ Types of jawfish--------(**4**)
- ✓ The Phanerozoic and the current extinction crises is sometime called....(**6th extinction**)
- ✓ Old bryozoans appeared during----------(**Cambrian**)
- ✓ The ---------- are called belly- footed mollusks _____?(**Gastropods**)
- ✓ Fossils in Fold belts are, except…… (**In Excellent Shape**).

Zoology Legends

- ✓ The order and succession of rocks is called (**lithostratograph**)
- ✓ The “belly-footed” mollusks___?(**Gastropods**)
- ✓ species of living insects -----?(**10 million**)
- ✓ Subphylum hexapoda example? (**Dragonflies**).
- ✓ Reptiles typically have ----bones in the lower jaw ? (**Six**)
- ✓ The brachiopod soft parts are enclosed by two morphologically different shells or valves with variety of muscles. ?
- ✓ (**Morphology**)
- ✓ *True multicellular are _ **eukaryotic**
- ✓ ***Wallace and Darwin** contribute in agy BRI statements thi
- ✓ *Octopus has _ **lost skeleton**
- ✓ *Modification of oparin_ **Haldane biochemical model**
- ✓ ***Archaeopteryx** is true link between _ reptiles and birds
- ✓ *Belly footed molusks_ **gastropods**
- ✓ *First vertebrate had_ **cartilaginous skeleton**
- ✓ *Water vascular system present in _ **echinoderms**

Zoology Legends

- ✓ Condon's commonly occurred as tooth-like elements that are useful in biostratigraphy, as are some other fish teeth and **scales** (ichthyoliths). Tetrapods
- ✓ "Which species of ape is closest to humans – the gorilla or chimp?" ✓ Putting together complete trees of all species ✓ Scala Nature: the ladder of life
- ✓ **Radiation Amniotes**: Late Carboniferous Three main clades.
- ✓ The three major skull patterns seen in amniotes: anapsid, diapsid and synapsid. (Based on Carroll 1987.)
- ✓ Early crocodilian largely terrestrial in habits walked on all fours and had an extensive armor of bony plates. More diverse and abundant during the **Jurassic and Cretaceous** than now.
- ✓ Progressionism **(Orthogenesis)**
- ✓ **Protozoa** Neither animal nor plant, but single-celled eukaryotes that show animal **characteristics** such as **motility and heterotrophy.**

Zoology Legends

- ✓ **Cnidarians:** The simplest of the higher metazoans with a radial diploblastic body plan and stinging cells or cnidoblasts. Sea anemones, jellyfish and hydra along with Corals
- ✓ **Cambrian Explosion**: The Cambrian explosion generated a range of new body plans during a relatively short time interval.
- ✓ Fossils in fold belts are except... **in excellent shape**
- ✓ Meaning of word "science" ... **Knowledge (**From Latin scientia, meaning "knowledge”**)**
- ✓ Chelicerates are diverse and heterogeneous includes...a) mites. b) Scorpian...c)spider d) all
- ✓ Water vascular system in ... **echinoderms**
- ✓ Lipids... Biomarkers, evidence for cyan bacteria and eukaryotes were around at least **2.7 Ga**
- ✓ **Georges Cuvier**: father of comparative anatomy
- ✓ Ancient life show era … **Paleozoic**
- ✓ Lithostratigraphic… **formation of rocks**

Zoology Legends

- ✓ **Finding the rhythm. Cyclostratigraphy** is the study of astronomically forced climate cycles within sedimentary successions.
- ✓ **Taphonomy** "is the paleontological study includes all the processes that occur after the death of an organism and before its fossilization in the rock".
- ✓ Current extinction crises called... **Sixth extinction**
- ✓ Alvarez.150KM crater, 10 KM Diameter of Meteorite 1991, Crater was identified in Chicxulub in Mexico
- ✓ **RNA world:** By fusion of organic molecules in the first billion years after the formation of the Earth. **Self-replicating RNA "RNA world".**
- ✓ **Hydrothermal Model:** Modification to the Oparin–Haldane biochemical model
- ✓ Diatom radiolarians... siliceous
- ✓ Siliceous **(diatoms, radiolarians)**
- ✓ **Evolutionary faunas modern developed during last ...550myr**

Zoology Legends

- ✓ Linguliformeans have: organophosphatic shells
- ✓ Which one is not property of mollusks… it is segmented
- ✓ Mass extinction **6 time**
- ✓ Life on Earth **4 Billion**
- ✓ Exoskeleton of arthropods made up of **chitin**
- ✓ Head fused with thorax **cephalothorax**
- ✓ Bivalved was named by **Linnaeus**
- ✓ Groups of conodont **3**
- ✓ Human arose **6-8mya**
- ✓ Insects first appeared early **devonian**
- ✓ Signor lipps effect **straightography**
- ✓ Bivalves is characterized by **(All)**
- ✓ Insect Population **10 million**
- ✓ Animal feed on died **plants and animals Scavenging**
- ✓ **Adapted** definition **Natural Selection**
- ✓ Structure and composition of fossil communities **Paleosynecology**

Zoology Legends

- ✓ Membrane enclose the embryo **Amnion**
- ✓ "Moss animals", **Filter feeders**, Sieve food particles using a retractable lophophore.
- ✓ Iso metric & allometric **growth.**
- ✓ The complete or partial remain of plant & animal body fossil
- ✓ Mass extinction on earth **6 Times**
- ✓ Crocodile more diverse in **Jurassic & Cretaceous**
- ✓ The Spiralians (mollusk–annelid–brachiopod–**bryozoans group**)
- ✓ Stenolaemata are a class of marine **bryozoans**.
- ✓ Palaeoendemics recently evolved **(neoendemics)**
- ✓ First appearance datum **(FAD)**
- ✓ **Thomas Kuhn:** science shuttles between so-called times of normal science and times of scientific revolution
- ✓ Echinoderms have **water vascular system**

Zoology Legends

- ✓ Snowball Earth by coincidence or design was a pivotal event in metazoan history; **bilaterians evolved after the Marinoan glaciation.**
- ✓ Isthmus of panama connect...**Connected North and South America**
- ✓ Largest **Arthropods Chelicerates**
- ✓ An example of a cnidarians is a **jellyfish.**
- ✓ **Sexual selection** is the set of evolutionary processes that depend on interactions between the sexes
- ✓ mollusks are typically encountered in various **aquatic environments**
- ✓ Reptiles typically have **six bones in the lower jaw**
- ✓ In mammals, there is a **single bone in the lower jaw,**
- ✓ **Who purpose natural selection? Charles Darwin**
- ✓ Echinoderms generally display **pentaradial symmetry**
- ✓ **(**The lower jaw of a **vertebrate consists of a single bone**, the mandible.)

Zoology Legends

- ✓ Humans and chimpanzees shared a common ancestor around **6-8 million** years ago
- ✓ (Crustaceans, such as crabs, lobsters, and shrimp, are characterized by the presence of claws.)
- ✓ The "**Signor-Lipps effect**" is related to: **Fossilization**
- ✓ **Abiogenesis** is the scientific term for the origin of life from non-living matter
- ✓ Physical processes that break fossils are: **Weathering and erosion**
- ✓ Fossils in fold belts are generally found in: **Poor shape**
- ✓ Who is considered the "Father of Anatomy"? **Galen**
- ✓ Natural selection definition: **Survival of the fittest**
- ✓ Filter feeders are characterized by: **Passive feeding**
- ✓ Formation of rocks is classified into: **Lithostratigraphic**
- ✓ What is a fossil? **Complete or partial remains of a plant or animal body**
- ✓ Palaeontology definition: **Study of ancient life forms**

Zoology Legends

- ✓ Ancient cultures often used: **Coral for medicine**
- ✓ Examples of Chelicerates are: **Spiders and scorpions**
- ✓ Arthropods have: **Segmented bodies and jointed appendages**
- ✓ Sexual selection in the evolutionary process refers to: **Competition for mates**
- ✓ Which of the following organisms exhibit a diploblastic body plan? **Cnidarians**
- ✓ Bryozoans are a group of: **Marine animals**
- ✓ Placedorm were: **Armored fish**
- ✓ Specialized technique commonly used to extract microfossils from rocks in petroleum exploration: **Palynology**
- ✓ When local ecological conditions affect the form of an organism during its life time and
- ✓ consequently fossil is called …(**Ecophenotypic Effects)**
- ✓ Jawless fishes that lived in ----------**period (Devonian)**
- ✓ Types of jawfish---**(4)**

Zoology Legends

- ✓ The Phanerozoic and the current extinction crises is sometime called (**6th extinction)**
- ✓ Old bryozoans appeared during---**(cambrain)**
- ✓ The ---------- are called belly- footed mollusks (Gastropods)
- ✓ Fossils in Fold belts are, except…… **(In Excellent Shape)**
- ✓ living insects -----? **(10 million)**
- ✓ Subphylum hexapoda example? (**Dragonflies**).
- ✓ Archaeopteryx is a true link between ……**reptiles and birds**
- ✓

ZOOLOGY
LEGENDS

Zoology Legends

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Remember me in you Precious Prayer

Jazak Allah